

Soft Jellyfish Propulsion Simulation Midterm Report

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Abstract—This report details progress made in the use of numerical methods for modeling soft robotic jellyfish propulsion and presents results obtained using a discrete elastic rod (DER) simulation. We implement a 2D, neutrally buoyant 5-node elastic beam with time-varying natural curvature to replicate jellyfish bell contractions, and we model hydrodynamic interactions using High-Reynolds-number Quadratic Drag Theory. With the quadratic drag theory implemented, the simulation now produces net forward motion. Additional testing on bell fineness ratio, contraction amplitude, and jetting contributions further demonstrate the sensitivity of jellyfish-inspired propulsion to morphology and actuation strategies.

I. PROBLEM STATEMENT AND MOTIVATION

Marine robots are becoming increasingly valuable to scientific study and commercial exploration, but remain limited in lifetime and range by propulsion sources that can be inefficient and hazardous to fragile sea-life [1]. As the field of soft robotics has matured, biomimetic designs including those inspired by jellyfish propulsion present a promising route to low-energy, highly maneuverable underwater platforms.

Jellyfish swim by producing coordinated bell contractions that generate vortex rings and jet thrust, achieving high efficiencies at low speeds [2]. This proposal seeks to evaluate the applicability of a discrete differential geometry model toward reproducing these fluid–structure interactions in simulation, which could decrease computational overhead for predicting the propulsive performance of soft robotic designs. Previous experimental and computational work on jellyfish-inspired robots and live-animal augmentation provides benchmarks and modeling motifs for our simulation-based exploration [4], [5].

II. INITIAL APPROACH

The jellyfish is modeled as a neutrally buoyant discrete elastic beam immersed in a viscous fluid. The simulation presented here implements a 2D beam consisting of five nodes connected by stretching springs and bending springs derived from DER energy formulations. Natural curvature is computed directly from the initial parabola-shaped bell. The initial parabolic nodal positions are defined parabolically according to Equation 1 for an n -node system.

$$\begin{cases} x(t) = wd * (\frac{t}{n-1} - \frac{1}{2}) \\ y(t) = -(ht * (\frac{2t}{n-1} - 1))^2 + y_{ctr} \end{cases} \quad t \in [0, n-1] \quad (1)$$

Time-varying curvature is then added as a periodic function of time to emulate the bell’s power stroke and recovery stroke. Curvature amplitude is highest near the center and

decreases toward the edges, with a delay term applied to capture the observed phase lag between central and peripheral contraction.

Hydrodynamic loading was initially modeled using low-Reynolds-number Resistive Force Theory (RFT), before switching to a high-Reynolds quadratic drag model. Drag forces proportional to the square of relative velocity are applied to each node [7]. These forces oppose motion but do not capture unsteady vortex generation or momentum jetting.

Time integration is performed using an implicit Newton–Raphson solver at each time step, combining inertial, elastic, viscous, and hydrodynamic contributions.

III. BACKGROUND AND EXISTING LITERATURE

A. Actuated Robotic Jellyfish: IPMC and SMA approaches

Najem et al. developed a biomimetic jellyfish-inspired underwater vehicle actuated by ionic polymer metal composites (IPMCs), demonstrating fabrication strategies and characterizing swimming performance for flexible-actuator-driven medusan robots. Their work shows that soft flexible actuators can reproduce bell-like contractions and serve as a practical benchmark for achievable kinematics in low-Reynolds number experimental platforms [2].

Earlier work at Virginia tech presented a jellyfish robot actuated by shape-memory-alloy (SMA) composite actuators. The robot, named “Robojelly” used SMA wires in parallel with spring steel spines to achieve large changes in natural curvature with magnitudes comparable to organic muscles. As shown in Figure 1, separate versions of the robot were built and characterized; one with a fully intact bell and one with “segmented” flaps around the bell margin. These impact of this change in bell shape could also be a target for modeling by our simulation. The study further emphasizes fabrication, kinematics replication, and experimental measurements of swimming performance, providing additional validation points for actuator timing and amplitude in biomimetic designs [1].

B. Mechanical Design and Motion Control

Although it revolves around a mostly rigid robot, the work presented at the Chinese Control Conference (Li & Yu) discusses mechanical structure-driven robotic jellyfish designs, motion analysis, and simulation of control schemes for steering and thrust modulation. An actuated ballast was incorporated to vary the center of the bell for bioinspired orientation control [3]. This study indicates the value of coupling mechanical design choices to control strategies to achieve desired swimming behaviors and maneuverability in robotic jellyfish.



Fig. 1: Images of Robojelly with unsegmented (top) and segmented (bottom) bell margin [1].

C. Computational FSI and Morphological Effects

Miles & Battista used IB2d, an open-source implementation of the immersed boundary mathematical framework, to study how tentacles and oral arms affect locomotion and found that appendages can strongly influence vortex formation and forward speed. In many parameter regimes, increased density or length of tentacles or oral arms were found to significantly decrease forward speed and change energy dissipation/mixing behavior [4].

D. Biological Augmentation and Application Context

Recent work by Anuszczyk & Dabiri demonstrates electromechanical stimulation of live jellyfish muscle tissue for ocean exploration applications, showing the potential for hybrid bio-robotic systems and reinforcing the broader interest in jellyfish-inspired platforms for low-power ocean monitoring shared in similar research by Xu et al. Their discussions provide context for realistic performance goals and operational use-cases for jellyfish-inspired vehicles [5], [6].

IV. SIMULATION MODEL IMPLEMENTATION

The implemented numerical simulator follows a discrete elastic rod formulation. Stretching forces and bending forces are computed from analytical gradients and Hessians of DER energy terms. Natural curvature is determined from the initial bell geometry using a discrete curvature operator based on adjacent tangent vectors.

A time-varying curvature function modulates this natural curvature, producing contraction and relaxation phases across each cycle. The activation pattern incorporates amplitude gradients, temporal delays, and exponential decay during recovery. It is defined as a piecewise function of cycle time by Equation 2.

$$\Delta\kappa(t) = \begin{cases} \Delta\kappa_{max} * \frac{t - t_{del}}{0.3P - t_{del}} & t \in [0, 0.3P) \\ \Delta\kappa_{max} * e^{-\frac{t - 0.3P}{\tau_{relax}}} & t \in [0.3P, P) \end{cases} \quad (2)$$

To move beyond the limitations of low-Reynolds-number resistive force theory, we incorporated a quadratic drag model of the form

$$\mathbf{F}_d = -\frac{1}{2}\rho C_D A \|\mathbf{v}_{rel}\| \mathbf{v}_{rel},$$

where C_D is a tunable drag coefficient [7]. Increasing C_D to values representative of high-Re flow enabled non-reciprocal momentum exchange with the surrounding fluid.

Newton–Raphson iteration was used to solve the implicit Euler method of integration for the nonlinear equations of motion at each time step, combining inertial, elastic, hydrodynamic, and external contributions.

V. SIMULATION PARAMETERS AND SETUP

The bell is initialized as a parabola spanning 0.1 m in width and 0.08 m in height. Five nodes discretize the geometry. Material and physical parameters include bending stiffness $EI = 0.01$, stretching stiffness $EA = 30$, RFT fluid drag coefficients $C_t = 0.15$ and $C_p = 0.30$, and a beam radius of 0.02 m with density 1050 kg/m³.

A high-Reynolds-number quadratic drag law was applied with heuristically determined drag coefficient $C_D = 100$ and seawater density of 1025 kg/m³. The timestep for implicit integration was $\Delta t = 0.01$ s.

Natural curvatures are computed from the resting shape, and time-dependent curvature is applied using a 4 s period, with a strong central contraction ($\Delta\kappa = -1.5$) and weaker edge actuation.

Outputs tracked include node positions, velocities, and lateral and vertical displacements of the end nodes and center node. Hydrodynamic forces on each node were also stored.

VI. RESULTS AND OBSERVATIONS

A. Bell Contraction and Advancement

The time variant natural curvature approach to modeling jellyfish contractions was successful in producing reciprocal motion of the bell. For each contraction cycle, the simulated jellyfish bell is initially parabolic in shape, but undergoes a shape change where natural curvature quickly increases to a peak value, resulting in the edges of the bell drawing closer together before slowly relaxing back to resting. This approach to kinematic modeling should provide a strong analogue for actuators used in jellyfish robots, like IPMC and BISMAL actuators, which also alter the bell’s natural curvature to produce motion [1], [2]. Images of the bell’s shape at various points within the contraction cycle are shown in Figure 2, with arrows at each node indicating hydrodynamic forces.

An early implementation attempted to model hydrodynamic forces in a low-Reynolds ($Re \ll 1$) environment by using resistive force theory (RFT) [8]. Although the vertical

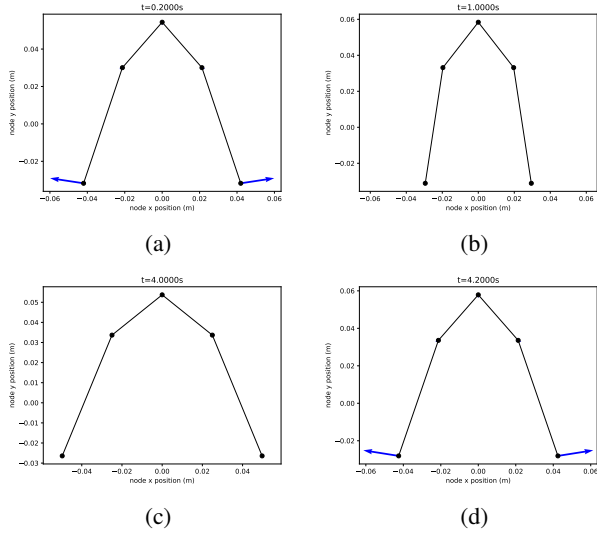


Fig. 2: Simulated bell shape at beginning of contraction (a), peak contraction (b), end of relaxation (c), and beginning of next cycle (d). Blue arrows indicate hydrodynamic drag forces acting on nodes.

positions of nodes shifted during the course of each individual cycle, this simulation failed to produce net movement of the structure over the full duration. This finding aligns with the scallop theorem, which explains that propulsion in low-Reynolds environments is not possible through geometrically reciprocal motion, even if the timescales are different for contraction and relaxation [9].

After further reading suggesting that natural jellyfish locomotion largely takes place at Reynolds numbers ranging from around 100 to 600, it became clear that even if it were to produce motion, RFT would have very limited applicability to the desired conditions, so a high-Reynolds quadratic drag model was instead implemented [10], [11]. The quadratic drag model initially also appeared to produce no forward motion when using a drag coefficient of 0.42 found in literature [6]. However, testing revealed that increasing C_D to 100 was able to produce realistic forward motion without fully preventing movement of the bell.

Figure 3 shows the vertical displacement of the simulated jellyfish compared to experimental data from a soft jellyfish robot built by Najem et al., showing relatively strong qualitative agreement with some experimental results [2].

The simulated results show a much higher amplitude of cyclic displacement per unit of forward progress compared to the experimental results, which advance more steadily with less backward motion per cycle. Although much of this disagreement can likely be attributed to shortcomings in the simulation, some of it may be a result of tracking the apex of the bell rather than its centroid. The jelly's centroid or any other node not at one of its extrema can be expected to show far less cyclic motion over the course of a contraction, thus producing a smoother displacement profile.

With the selected simulation parameters, the jellyfish bell moved an average of approximately 0.0037 meters per contraction, giving it an average upward velocity of 0.0009

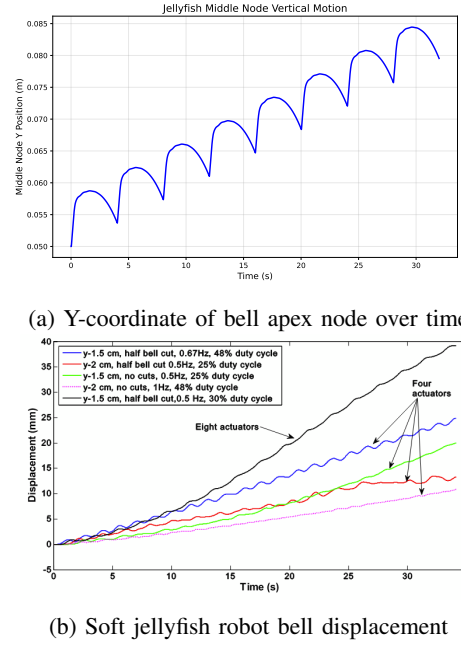


Fig. 3: Comparison of vertical displacement plots between simulation (a) and experimental data (b) [2]

m/s. This average velocity falls within the experimentally observed velocities of Najem et al.'s robot, which range from 0.0003 to 0.001 m/s. However, the experimental robot had a much faster contraction frequency of 0.5-1 Hz, compared to our frequency of 0.25 Hz, and also had a smaller bell size [2].

In addition to the vertical position of the apex node, horizontal positions of the bell's edges were also tracked during the simulation. Figure 4 plots the x-coordinates of the first and last nodes, which converge toward zero in each cycle as the curvature rises before gradually relaxing back to their starting points. Also visible in these plots is how changes in the direction of travel appear to generate oscillations in position which are not present in real world jellyfish or jellyfish robots, suggesting the damping ratio of our simulation was less than that of real world bells due to some combination of inaccuracies in material properties, hydrodynamic forces, and nodal mass.

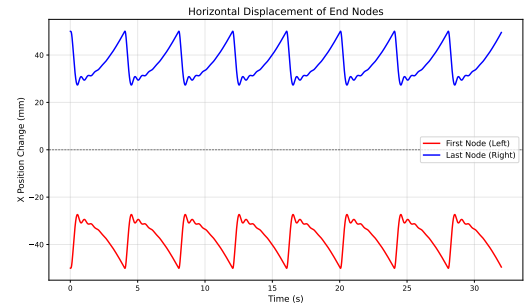


Fig. 4: Horizontal displacement trajectories of the system's end nodes, representing the edges of the jelly's bell.

B. Varying Fineness Ratio

Bell geometry is a major factor in the performance and optimal control strategy of both real jellyfish and bioinspired medusa robots. Biological jelly species are commonly differentiated according to their fineness ratio, which Equation 3 defines as the ratio of bell height to width (or diameter in 3D) [2], [11].

$$F = h/d \quad (3)$$

Jellyfish with high fineness ratio (often defined as $F > 0.5$ or $F > 1$) are known as prolate species, and can be characterized by fast, jetting dominated movement with lower efficiency and more uniform bell contraction. Jellyfish with low fineness ratio (usually $F < 0.5$) are known as oblate species, and can be characterized by slow, vortex dominated movement with higher efficiency and cell contraction localized mostly to the edges [2], [11].

To compare the effects of bell geometry in simulation, trials were run with bell fineness ratios ranging from 0.4 to 1. All simulations were run with the same starting position, curvature change, and cycle period of 3 seconds. Figure 5 displays the bell shape of all 4 trials after 3 cycles have elapsed.

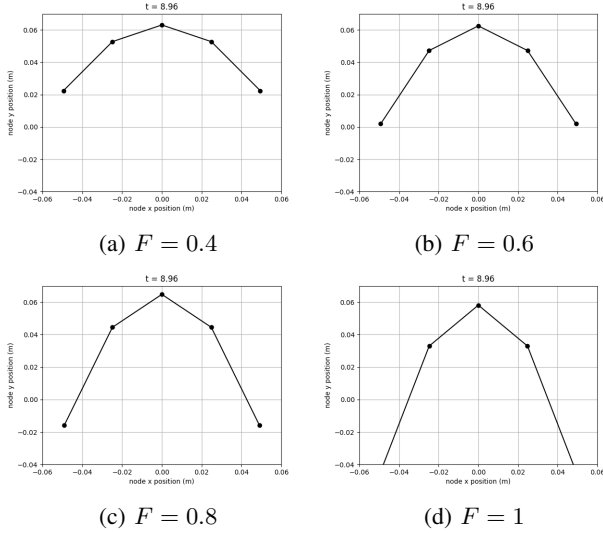


Fig. 5: Simulated bell shape in undeformed configuration after 3 cycles for a range of fineness ratios

The three simulations with fineness ratio < 1 all show similar displacements, with the $F = 0.8$ bell having moved the farthest, while the $F = 1$ bell has much less displacement. These findings seemingly indicate that with all other parameters held constant the optimal fineness ratio is closest to 0.8. Because the simulation does not model the effects of jetting or vortices, bells with more extreme fineness ratios which would normally benefit from either of these phenomena are artificially penalized.

C. Varying Contraction Depth and Speed

The magnitude of bell contraction is one control parameter that creates an opportunity for optimization. Higher bell

contraction results in a smaller final bell volume and thinner hydrodynamic profile with less frontal area to create pressure drag, factors that should aid in propulsion. However, it also requires that the actuators exert more work to deform a greater distance. In a soft robot modeled as an elastic shell with open resting shape, greater deflection can also be expected to require more force unless the force profile is designed to be highly nonlinear. Finally, a robot capable of deep contraction faces additional design challenges to enable its greater range of motion, especially for the robot actuators that have been previously used like BISMAs and IPMCs [1], [2].

To briefly assess the impact of bell contraction depth on propulsion, the jellyfish bell was simulated with two different activation functions, one which applies a small curvature change to exclusively the apex node and another that applies large curvature changes to all interior nodes. Figure 6 shows the state of the higher curvature bell at two points within its cycle, one where it has reached maximum curvature and one where hydrodynamic forces are acting in a net downward direction. These results reveal an additional unforeseen cost to high contraction depth: when end nodes start to swing upward they encounter a net downward drag force that prevents the jelly from coasting upward. Furthermore, as the bell edges become more vertical the downward component of drag becomes smaller and less forward work is done. Based on these results it appears that any contraction where the edges reach vertical is wasting energy and thus less optimal.

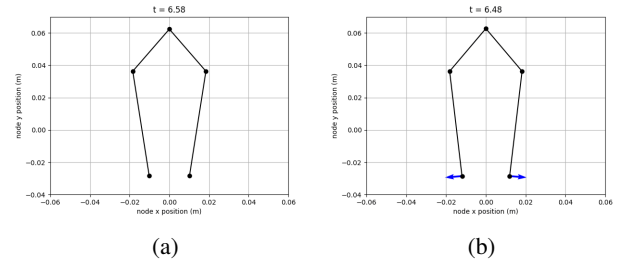


Fig. 6: High curvature jellyfish bell max contraction (a) and downward force (b) states

Another important control parameter is the frequency of contraction. The initially tested period of 4 seconds was mostly arbitrary, so another simulation was run using a cycle period of 2 seconds, giving a contraction frequency of 0.5 Hz. Figure 7 shows the resulting displacement profile, in which velocity has increased to about 0.0014 m/s but distance traveled has decreased to about 0.003 meters per cycle.

Doubling the contraction frequency has the downside of more than doubling power consumption, because each cycle costs more energy and twice as many cycles are needed. Because the distance traveled increased by less than 100% the cost of transport must be higher. From these results the optimal contraction frequency for the design in question is less than 0.5 Hz, but may be either greater or less than 0.25 Hz.

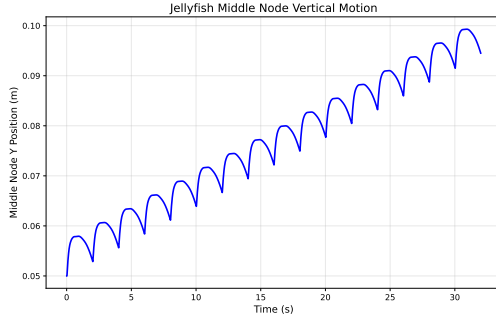


Fig. 7: Vertical displacement of the jellyfish for higher contraction frequency

D. Simplified Jetting Model

A simplified approximation of jetting thrust was implemented for some simulated by adding a distributed external force based on the conservation of linear momentum for a fixed control volume. Equation 4 was used to model this force, where the rate of change of volume under the bell was found by calculating the enclosed area at the previous and current timestep.

$$F = \rho \vec{V} \frac{dV}{dt} \quad (4)$$

Several simulations were run with this additional jetting force included, but its utility is highly limited by the unreliability of the assumptions made in its derivation, including the assumption that the control volume does not deform or accelerate, and the application of calculated force as a uniform load distributed equally across all nodes. Compared to simulations with only drag forces, simulations with jetting thrust included resulted in upward of a 20% increase in distance traveled, with increases strongly dependent on geometric parameters like bell width.

VII. DISCUSSION

The emergence of forward propulsion following the introduction of quadratic drag highlights the importance of operating in a high-Reynolds-number regime, where asymmetric momentum exchange can occur even with relatively simple kinematics. The sensitivity of thrust to bell fineness ratio aligns with biological categorizations of oblate and prolate medusae, reinforcing the role of geometry in jetting and vortex-ring formation [2].

Contraction amplitude testing revealed that greater deformation does not necessarily increase performance, as excessive lateral drag can overwhelm gains in jetting thrust. The simplified jetting model further demonstrated that even coarse approximations of momentum expulsion substantially influence net displacement, suggesting that more sophisticated vortex-based hydrodynamic modeling could significantly enhance accuracy.

VIII. LIMITATIONS

Although the final model produced forward propulsion, it remains limited by several simplifying assumptions. The

quadratic drag model captures high-Re behavior only in certain Reynolds number ranges, without reproducing vortex-ring dynamics that dominate real jellyfish propulsion[9]. The jetting term is based solely on area change and does not explicitly simulate interior fluid mass or vortex shedding [2], [5].

The structure is discretized with only five nodes, restricting the model's ability to capture traveling-wave deformations. Finally, the 2D formulation omits out-of-plane effects, preventing accurate modeling of bell curvature distribution and three-dimensional flow structures.

IX. FUTURE WORK

Future improvements include developing a more complete hydrodynamic model that includes jetting, vortex formation, and more complex high-re behaviors. This along with implementing the model in 3D using a shell simulation would allow for far more accurate modeling and results.

X. CONCLUSION

We implemented a discrete elastic beam simulation with time-varying natural curvature to model jellyfish bell contractions. The framework successfully reproduces cyclic bell deformation, and due to the high-Re hydrodynamic model, generates vertical displacement corresponding to the expected vertical motion from the time changing natural curvature. The final simulation enabled exploration of key morphological and actuation parameters, including fineness ratio and contraction amplitude, and demonstrated trends consistent with biological medusan swimming.

Although simplified, the model establishes a computationally efficient foundation for simulating soft aquatic robot propulsion and highlights clear pathways toward more realistic fluid–structure interaction modeling.

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